

KCPC $\times$ KCL WiSTEM

Problem Sheet

Logic & Lattes

Developed by the KCPC Internal Committee, this problem sheet accompanies the joint *KCL WiSTEM* social workshop. Problems are divided into three categories:

- Logical Problems
 - Programming Problems
 - Extended Problems
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I. Logical Problems

1. **Left-Handed People**

In a room of 100 people, 99% are left-handed.

Question. How many people must leave the room in order for the percentage of left-handed people to decrease to 98%?

2. **The Three Wise Men**

A king called the three wisest men to his court to decide who would become his new advisor. He placed a hat on each man's head such that each man could see the other two hats, but not his own. Each hat was either white or blue. The king guaranteed that at least one person was wearing a blue hat. The contest was declared fair to all three men, and they were forbidden from speaking.

After sitting in silence for a long time, one man stood up and correctly announced the color of his own hat.

Question. What color did he announce, and how did he deduce it?

3. **Ants on a Ruler**

One hundred ants are dropped simultaneously at arbitrary positions on a straight ruler of fixed length L . Each ant walks at a constant speed v , choosing an initial direction (left or right) at random. Whenever two ants collide, they instantly reverse directions and continue at the same speed v . Ants fall off the ruler as soon as they reach either end.

Question. Determine the time T one must wait to be certain that all ants have fallen off the ruler, regardless of their initial positions and directions.

4. **Simple Stick Game**

Consider a game in which there is initially a heap of n sticks. Players A and B move alternately, and player A begins. On each move, the player must remove 1, 2, or 3 sticks from the heap. The player who removes the last stick wins the game.

Question. Determine which player has a winning strategy in terms of the initial value of n .

5. **Highly Divisible Triangular Number**

The sequence of triangular numbers is generated by summing the natural numbers. Thus the 7th triangular number is

$$1 + 2 + 3 + 4 + 5 + 6 + 7 = 28.$$

The first ten triangular numbers are

$$1, 3, 6, 10, 15, 21, 28, 36, 45, 55, \dots$$

Let us list the factors of the first seven triangular numbers:

$$\begin{aligned} \mathbf{1} &: 1 \\ \mathbf{3} &: 1, 3 \\ \mathbf{6} &: 1, 2, 3, 6 \\ \mathbf{10} &: 1, 2, 5, 10 \\ \mathbf{15} &: 1, 3, 5, 15 \\ \mathbf{21} &: 1, 3, 7, 21 \\ \mathbf{28} &: 1, 2, 4, 7, 14, 28 \end{aligned}$$

We can see that 28 is the first triangular number to have more than five divisors.

Question. What is the value of the first triangular number to have more than 500 divisors?

6. **Card Sums Puzzle**

Five friends are each dealt two cards from a set of twelve cards numbered 1 to 12 (inclusive). Each card is used exactly once.

In turn, the friends announce the sum of the values of their two cards:

- Paolo scores 4
- Quinn scores 11
- Romy scores 16
- Stephen scores 19
- Thomas scores 20

Question. Which of the following statements is true?

- A. Paolo has card 2
- B. Quinn has card 3
- C. Romy has card 5
- D. Stephen has card 7
- E. Thomas has card 11

7. **Aardvarks and Anteaters**

Together, n aardvarks and 12 anteaters eat $n^2 + 20n + 25$ ants. Each animal eats the same whole number of ants.

Question. How many ants does each animal eat?

- A. 61
- B. 66
- C. 71
- D. 74
- E. 79

8. **Shaded Squares**

Fifty squares are drawn side by side in a line. The first and last squares are shaded. Other squares in the line must be shaded such that both of the following rules apply:

- (a) No two adjacent squares are shaded.
- (b) There are no more than three consecutive unshaded squares.

Question. What is the difference between the smallest and largest number of squares that can be shaded?

- A. 8
 - B. 10
 - C. 11
 - D. 13
 - E. 15
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II. Programming Problems

These problems require programming knowledge and careful modelling!

1. LeetCode 11 - Container With Most Water

(This problem makes use of the array data structure alongside what is known as a "Two Pointer" approach.)

<https://leetcode.com/problems/container-with-most-water>

2. LeetCode 7 - Reverse Integer

(Be very careful when reading this problem description! It's a bit tricky, but that's exactly why we want to do it :D)

<https://leetcode.com/problems/reverse-integer>

3. LeetCode 13 - Roman to Integer

(This problem makes use of implicitly storing values and mapping them, imagine the concept of using a Dictionary for word lookup (but mathematical!))

<https://leetcode.com/problems/roman-to-integer>

4. LeetCode 258 - Add Digits

(This problem makes use of number theory! It's a foundation point when it comes to the digital root pattern.)

<https://leetcode.com/problems/add-digits>

III. Extended Problems

Important. The following problems are genuinely extremely difficult. They are intended as stretch challenges for those who wish to explore advanced modelling and optimisation techniques.

Matthew's Note. I hope you enjoy working through these problems. If you have any questions, please feel free to contact the KCPC Director of Competitive Programmes (Marcell Nemeth) or myself (Matthew)!

1. LeetCode 1928 - Minimum Cost to Reach Destination in Time

(State expansion + constrained shortest path)

<https://leetcode.com/problems/minimum-cost-to-reach-destination-in-time>

2. **Codeforces 786B - Legacy**
(Segment tree graph construction + Dijkstra)
<https://codeforces.com/problemset/problem/786/B>
3. **AtCoder ABC 213 E - Stronger Takahashi**
(Advanced 0–1 BFS modelling; non-trivial movement rules)
https://atcoder.jp/contests/abc213/tasks/abc213_e
4. **LeetCode 23 - Merge K Sorted Lists**
(Heaps; Linked Lists; Divide & Conquer)
<https://leetcode.com/problems/merge-k-sorted-lists>
5. **LeetCode 4 - Median of Two Sorted Arrays**
(Arrays; Binary Search; Divide & Conquer)
<https://leetcode.com/problems/median-of-two-sorted-arrays>
6. **LeetCode 87 - Scramble String**
(Strings; Dynamic Programming)
<https://leetcode.com/problems/scramble-string>

Still interested in checking out more problems? Feel free to take a look through all of our problem sets :D They're available on both our Discord and the KCPC website!